MILITARY SPECIFICATION

BOOSTER ASSEMBLY FOR DEMOLITION KIT CRATERING, M180 PARTS FOR, LOADING, ASSEMBLING & PACKING

This specification is approved for use by US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains the parts, loading assembly, and packing of one type of Booster Assembly for one type of rocket motor assembly designated as Rocket Motor for Demolition Kit, Cratering M180.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents.-The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-P-48395 - Pellets, Explosive Compositions

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

DRAWINGS

US ARMY RESEARCH AND DEVELOPMENT COMMAND

9244452 - Booster Assembly

FSC: 1340

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-199.

(The Code of Federal Regulations is available from Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders shall specify 49CFR 100-199 (latest revision).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Components and assembly.- The components and assembly shall comply with all requirements specified on Dwg. 9244452 and associated drawings, and with all requirements specified in applicable specification and standards.

3.2.1 Pellets.-The pellets shall comply with all requirements specified in MIL-P-48395 - Pellets, Explosive Compositions.

3.3 Explosive Lead Density Test.-Composition A5 in the explosive lead (see Dwg. 9312799) shall meet the specific gravity when tested as specified in 4.5.2.

3.4 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. All packing components shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Base, Booster	9244448	10
Cup, Booster	9244450	10
Cup and Base Assembly	9244451	10
Booster Assembly	9244452	10
Washer, Retainer	9244453	10
Disc, Booster	9244454	10
Lead, Explosive PA503	9312799	10
Cup, Lead	9312808	10

4.3.2 Inspections to be performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63086 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Booster Assembly				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Base, Booster (Dwg. 9244448) Examination for Defects	10		3.2	4.4.2.1
	Cup, Booster (Dwg. 9244450) Examination for Defects	10		3.2	4.4.2.2
	Cup and Base Assembly (Dwg. 9244451) Examination for Defects	10		3.2	4.4.2.3
	Booster Assembly (Dwg. 9244452) Examination for Defects	10		3.2	4.4.2.4
	Washer, Retainer (Dwg. 9244453) Examination for Defects	10		3.2	4.4.2.5
	Disc, Booster (Dwg. 9244454) Examination for Defects	10		3.2	4.4.2.6
	Lead, Explosive PA503 (Dwg. 9312799) Examination for Defects	10		3.2	4.4.2.7
	Explosive Lead Test	10		3.3	4.5.2
	Cup, Lead (Dwg. 9312808) Examination for Defects	10		3.2	4.4.2.8

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Booster Assemblies shall contain:

a. Metal parts and assemblies from not more than one interfix lot number from one manufacturer.

b. Explosive lead assemblies from not more than one interfix lot number from one manufacturer.

c. Composition A5 from not more than one interfix lot number from one manufacturer.

d. Booster pellets from not more than one interfix lot number from one manufacturer.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plans - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63086 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Base, Booster		9244448
			NEXT HIGHER ASSEMBLY 9244451

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length		0.40%	3.2	Gage
102	True position of bottom hole with datum A.		0.40%	3.2	Gage
103	Outside diameter		0.40%	3.2	Gage
Minor					
201	Location of small hole		0.65%	3.2	Gage
202	Diameter of small hole		0.65%	3.2	Gage
203	Thickness		0.65%	3.2	Gage
204	Diameter of bottom hole		0.65%	3.2	Gage
205	Radius at bottom, max.		0.65%	3.2	Gage
206	Cut out missing at top		0.65%	3.2	Visual
207	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63086 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9244450
4.4.2.2	Cup, Booster				NEXT HIGHER ASSEMBLY 9244451
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length		0.40%	3.2	Gage
102	True position of small hole with datum A		0.40%	3.2	Gage
103	Inside diameter		0.40%	3.2	Gage
Minor					
201	Diameter of small hole		0.65%	3.2	Gage
202	Thickness of bottom		0.65%	3.2	Gage
203	Outside diameter		0.65%	3.2	Gage
204	Radius at bottom, max.		0.65%	3.2	Gage
205	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63086 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Cup and Base Assembly		9244451
			NEXT HIGHER ASSEMBLY 9244452

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length, max.		0.40%	3.2	Gage
102	True position of large hole with datum A.		0.40%	3.2	Gage
103	Brazing improper		0.40%	3.2	Visual
Minor					
201	Flatness of Base		0.65%	3.2	Gage
202	Protective finish inadequate, scratched or damaged		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Booster Assembly		9244452
			NEXT HIGHER ASSEMBLY
			9244401

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Total length, maximum (max.)		0.40%	3.2	Gage
102	Swage missing, improper, or less than 360°		0.40%	3.2	Visual
103	Any part missing or not assembled properly		0.40%	3.2	Visual
104	Adhesive missing		0.40%	3.2	Visual
105	Presence of explosive on outside of booster cup		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63086 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Washer, Retainer		9244453
			NEXT HIGHER ASSEMBLY 9244452

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
102	Outside diameter		0.40%	3.2	Gage
103	Diameter of hole		0.40%	3.2	Gage
Minor					
201	Thickness		0.65%	3.2	Gage
202	Protective finish inadequate, scratched or damaged		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B -63086 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Disc, Booster		9244454
			NEXT HIGHER ASSEMBLY
			9244452

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
103	Flatness		0.40%	3.2	Gage
Minor					
201	Protective finish inadequate, scratched or damaged		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63086(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Lead, Explosive PA503		9312799
			NEXT HIGHER ASSEMBLY
			9244452

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of lead cup, max.		0.40%	3.2	Scale
102	Height of cup from flange, max.		0.40%	3.2	Gage
103	Depth of charge		0.40%	3.2	Gage
104	Explosive lead test		0.10%	3.3	4.5.2
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63086 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Cup, Lead		9312808
			NEXT HIGHER ASSEMBLY 9312799

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Flange diameter, max.		0.40%	3.2	Gage
102	Length from bottom of cup to flange, max.		0.40%	3.2	Gage
103	Outside diameter		0.40%	3.2	Gage
Minor					
201	Thickness of bottom, min.		0.65%	3.2	Gage
202	Radius at bottom		0.65%	3.2	Gage
203	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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4.4.3 Testing.-Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment.-The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test method and procedures

4.5.1 Materials, components and processes.-Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections.-The visual, manual and mechanical inspections and tests shall verify compliance with requirements of sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Explosive Lead Test.-The explosive lead shall be weighed on a balance and the weight recorded as dry weight. The lead shall then be weighed while completely submerged in water and the weight recorded as weight under water. After both these operations have been completed, the Comp A5 shall be dissolved from the assembly using warm cyclohexanone. Then the lead cup shall be weighed on a balance and the weight recorded as dry weight. Then the lead cup shall be weighed while completely submerged in water and the weight recorded as weight under water. The temperature of the water shall be 68 degrees F plus or minus 5 degrees F. Specific gravity of the Comp A5 shall be calculated as follows:

$$\text{Specific Gravity} = \frac{A-C}{(A-C) - (B-D)}$$

Where:

A = Dry weight of assembly
B = Weight under water of assembly
C = Dry weight of cup only
D = Weight under water of cup only

5. PACKAGING

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5.1 Preservation and packaging

5.1.1 Level C.-The booster assembly shall be packaged in accordance with Code of Federal Regulations 49 CFR parts 171-179.

5.2 Packing

5.2.1 Level C.-The booster assembly shall be packed in accordance with Code of Federal Regulations 49 CFR parts 171-179 and in a manner that will assure safe delivery from the manufacturer's plant to the first receiving activity.

5.3 Marking.-The unit packages shall be marked in accordance with DOT Regulations and MIL-STD-129.

6. NOTES

6.1 Intended use.-The booster assembly covered by this specification is used in the Rocket Motor Assembly and is a component of the Demolition Kit Cratering M180.

6.2 Ordering data.-See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval.-See MIL-A-48078. Submit equipment designs as required to Commander U.S. Army Research and Development Command, ATTN: DRDAR-QAR-I, Dover, N.J. 07801.

6.4 Drawings.-Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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